

A 3.3.4 SQL Aggregate Functions (HL Only)



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SQL aggregate functions (AVG, COUNT, MAX, MIN, SUM) perform calculations on grouped data for reporting and decision-making within a database. (HL only)

Data Aggregation

In many real-world scenarios, you won't just need to retrieve individual records from a database. The most valuable information often comes from analysing trends, summarising data, and performing calculations across multiple records with certain characteristics. This is where aggregate functions come into play.

Aggregate functions in SQL allow you to perform calculations on a set of rows and return a single, summarised value. These functions are indispensable for generating reports that provide an overview of your data and for making data-driven decisions. When combined with the **GROUP BY** clause, their power is amplified, enabling you to perform these calculations on specific subsets of your data.

Introducing the Key Aggregate Commands

The SQL standard provides several built-in aggregate functions that you must understand and be able to construct. Let's explore each of them in detail:

1. AVERAGE (AVG)

The **AVG()** function calculates the arithmetic mean (average) of a set of numeric values in a specified column.

Syntax:

```
SELECT AVG(column_name)
FROM table_name
WHERE condition;
```

Explanation:

- **AVG(column_name)**: This specifies the column containing the numeric values for which you want to calculate the average.
- **FROM table_name**: This indicates the table from which you are retrieving the data.
- **WHERE condition**: (Optional) This clause allows you to filter the rows before calculating the average.

Example for Reporting:

Imagine you have a table named **Sales** with columns **ProductID** and **Price**. To find the average price of all products sold, you would use:

```
SELECT AVG(Price)
FROM Sales;
```

Example for Decision-Making with Grouping:

Now, let's say your **Sales** table also has a **Category** column. To determine the average price of products within each category, which could inform pricing strategies, you would use the **GROUP BY** clause:

```
SELECT Category, AVG(Price)
FROM Sales
GROUP BY Category;
```

This query will return a result set where each row shows a category and the average price of products in that category.

2. COUNT

The **COUNT()** function determines the number of rows that meet a specific criterion. It has a couple of variations:

- **COUNT(*):** Counts all rows in a table or a group, regardless of whether the columns contain NULL values.
- **COUNT(column_name):** Counts the number of rows in a specified column with non-NULL values.

Syntax:

```
SELECT COUNT(*) | COUNT(column_name)
FROM table_name
WHERE condition;
```

Explanation:

- **COUNT(*)** or **COUNT(column_name):** Specifies what you want to count.
- **FROM table_name:** The table from which to retrieve data.
- **WHERE condition:** (Optional) Filters rows before counting.

Example for Reporting:

To find the total number of sales records in the **Sales** table:

```
SELECT COUNT(*)
FROM Sales;
```

To find the number of sales records that have a recorded **CustomerID**:

```
SELECT COUNT(CustomerID)
FROM Sales;
```

Example of Decision-Making with Grouping:

To find the number of products sold in each **Category** in the **Sales** table:

```
SELECT Category, COUNT(*)
FROM Sales
GROUP BY Category;
```

This would provide insights into different product categories' popularity or sales volume.

3. MAX

The `MAX()` function retrieves the highest value from a set of values in a specified column. This can be used for numeric, string (lexicographically), or date/time data types.

Syntax:

```
SELECT MAX(column_name)
FROM table_name
WHERE condition;
```

Explanation:

- `MAX(column_name)`: The column from which to find the maximum value.
- `FROM table_name`: The table to query.
- `WHERE condition`: (Optional) Filters rows before finding the maximum.

Example for Reporting:

To find the highest price of any product in the `Products` table:

```
SELECT MAX(Price)
FROM Products;
```

Example for Decision-Making with Grouping:

To find the most expensive product in each `Category`:

```
SELECT Category, MAX(Price)
FROM Products
GROUP BY Category;
```

This information could be useful for understanding the premium offerings within each category.

4. MIN

Conversely, the `MIN()` function retrieves the lowest value from a set of values in a specified column. Similar to `MAX()`, it can be used with numeric, string, and date/time data types.

Syntax:

```
SELECT MIN(column_name)
FROM table_name
WHERE condition;
```

Explanation:

- **MIN(column_name)**: The column from which to find the minimum value.
- **FROM table_name**: The table to query.
- **WHERE condition**: (Optional) Filters rows.

Example for Reporting:

To find the lowest price of any product in the **Products** table:

```
SELECT MIN(Price)
FROM Products;
```

Example for Decision-Making with Grouping:

To identify the least expensive product in each **Category**:

```
SELECT Category, MIN(Price)
FROM Products
GROUP BY Category;
```

This could be important for understanding each category's entry-level products or price floor.

5. SUM

The **SUM()** function calculates the total sum of numeric values in a specified column.

Syntax:

```
SELECT SUM(column_name)
FROM table_name
WHERE condition;
```

Explanation:

- `SUM(column_name)`: The column containing the numeric values to be summed.
- `FROM table_name`: The table to query.
- `WHERE condition`: (Optional) Filters rows before summation.

Example for Reporting:

To calculate the total revenue from all sales in the `Sales` table (assuming a `Revenue` column):

```
SELECT SUM(Revenue)
FROM Sales;
```

Example of Decision-Making with Grouping:

To calculate the total revenue generated by each `Category`:

```
SELECT Category, SUM(Revenue)
FROM Sales
GROUP BY Category;
```

This provides a clear picture of which product categories are contributing the most to the overall revenue, aiding strategic decision-making.

The Synergy of Aggregate Functions and GROUP BY

As you've seen in the examples, the `GROUP BY` clause is often used in conjunction with aggregate functions. The `GROUP BY` clause groups rows with the same values in one or more specified columns into a summary row. Aggregate functions are then applied to each of these groups independently.

General Syntax with GROUP BY:

```
SELECT column1, column2, ..., aggregate_function(columnX)
FROM table_name
WHERE condition
GROUP BY column1, column2, ...
ORDER BY column1, column2, ...;
```

Key Points about GROUP BY:

- Any column listed in the **SELECT** statement that is *not* an aggregate function must be included in the **GROUP BY** clause. This ensures that the results are logically grouped.
- The **GROUP BY** clause comes after the **WHERE** clause (if present) and before the **ORDER BY** clause (if present).
- You can group by multiple columns, creating finer levels of aggregation. For example, you could group sales by **Category** and then by **Month** to analyse monthly performance within each category.

Example: Analysing Student Grades

Consider a table named **Grades** with columns **StudentID**, **Course**, and **Score**. To find the average score for each course:

```
SELECT Course, AVG(Score)
FROM Grades
GROUP BY Course;
```

This would give you a report showing the average performance of students in each individual course, valuable for curriculum evaluation.

Practical Applications for Reporting and Decision-Making

The ability to construct calculations with SQL aggregate functions on grouped data opens up a wide range of possibilities for reporting and informed decision-making:

- **Sales Analysis:** Calculating total sales, average order value, number of customers, or best-selling products by region, month, or customer segment. This helps in identifying trends, understanding customer behaviour, and optimising sales strategies.
- **Financial Reporting:** Calculating total revenue, expenses, or profits by department, quarter, or product line. This is crucial for understanding financial performance and making strategic financial decisions.
- **Inventory Management:** Determining the total quantity of each product in stock, the average lead time for restocking, or the minimum and maximum stock levels. This aids in efficient inventory control and prevents stockouts or overstocking.
- **Website Analytics:** Calculating the number of visitors, average session duration, or bounce rate for different pages or traffic sources. This provides insights into user engagement and helps in optimising website design and content.

- **Educational Performance:** Analysing average scores, pass rates, or the number of students in different courses or grade levels. This helps in evaluating teaching effectiveness and identifying areas for improvement.
- **Resource Management:** Calculating the total usage of resources (e.g., equipment, rooms) by department or time period. This aids in efficient allocation and planning of resources.

Important Considerations

- **NULL Values:** Aggregate functions generally ignore NULL values in calculations, except for `COUNT(*)`, which counts all rows regardless of NULLs. Be mindful of how NULL values might affect your results.
- **Data Types:** Ensure that the columns you are applying aggregate functions to have appropriate data types (e.g., numeric for `AVG`, `SUM`, `MIN`, `MAX`). Applying these functions to non-numeric data may lead to errors or unexpected results.
- **Performance:** On very large datasets, using aggregate functions with `GROUP BY` can be resource-intensive. Ensure your database is properly indexed to optimise query performance.
- **The HAVING Clause:** If you need to filter the results of a grouped query based on the aggregate function's value, you use the `HAVING` clause instead of `WHERE`. The `HAVING` clause filters groups after the aggregation has been performed.

```
SELECT Category, AVG(Price) AS AveragePrice
FROM Sales
GROUP BY Category
HAVING AVG(Price) > 50; -- Only show categories with an average price greater
```



Practice Questions

1. Using a table named `Orders` with columns `OrderID`, `CustomerID`, and `OrderAmount`, write an SQL query to find the total order amount for each customer.
2. Using a table named `Products` with columns `ProductID`, `Category`, and `StockQuantity`, write an SQL query to find the category with the fewest products in stock.
3. Using a table named `ExamScores` with columns `StudentID`, `Subject`, and `Score`, write an SQL query to find the highest score achieved in each subject.

4. Using a table named `WebsiteVisits` with columns `VisitID`, `PageURL`, and `VisitDate`, write an SQL query to find the number of visits to each unique `PageURL`.
5. Using a table named `Employees` with columns `EmployeeID`, `Department`, and `Salary`, write an SQL query to find the average salary for each department, only showing departments with more than 5 employees.

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